MATH 3042- Lecture 3- Filled In

January 23, 2026 10:19 AM

Unit 3- Probability Rules for Computing[1]

Basic Probability Foundations

Sample Spaces and Events

Definition: A sample space S is the set of all possible equally likely outcomes of an experiment. An event A is a subset of the sample space.

Example 1 - Network Packets: A router receives packets that can be classified by priority (High, Medium, Low) and protocol (TCP, UDP). $3 \times 2 = 6$

- Sample Space: S = {(H,TCP), (H,UDP), (M,TCP), (M,UDP), (L,TCP), (L,UDP)}
- |S| = 6 ✓
- Event A = "High priority packet" = {(H,TCP), (H,UDP)} $P(A) = \frac{2}{6}$
- Event B = "TCP packet" = {(H,TCP), (M,TCP), (L,TCP)} $P(B) = \frac{3}{6}$

Basic Probability: For equally likely outcomes, $P(A) = \frac{|A|}{|S|}$

Example 2 - Bit Errors: In a 4-bit transmission, what's the probability that exactly 2 bits are corrupted? Assume that the probability of a corrupted bit is $\boxed{10\%}$

R = Right/Correct
W = wrong/corrupted

All possibilities of W's and R's = $\underline{2}\ \underline{2}\ \underline{2}\ \underline{2} = 2^4 = 16$

- Patterns with exactly 2 errors:
- P(exactly 2 errors) =

Exactly 2 R 2 W: $\frac{4!}{2!2!} = C(4,2) = 6$

RRWW
RWRW
WRWR
WWRR
RWWR
WRRW

$6 \times (0.9)^2 (0.1)^2$

Complement Rule

$$\boxed{\textbf{Rule}: P(A') = 1 - P(A)}$$

A' = not A

This is especially useful when counting the complement is easier than counting the event itself.

Example - System Availability: For a system with two independent servers, if the probability that a server is down is 0.02, then:

P(at least one server is up) = 1 – P(both servers are down) = $1 - 0.0004 = 99.96\%$

0.02×0.02

[1] I have used Claude Sonnet 4.5 to edit, structure and generate some of the examples in these lecture notes

Example - Password Security: A password must contain at least one special character. If there are 95 printable ASCII characters and 26 of them are special characters: not special = 95 − 26 = 69

Instead of counting passwords with one or mores special character directly, count passwords with no special characters and use the complement rule

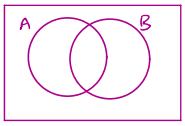
For a 6-character password: P(at least one special) = 1 − P(no special characters)

$$1 - \frac{69^6}{95^6} = 85.3\%$$

Addition and Multiplication Rules

Rule of Sum (Addition Rule)

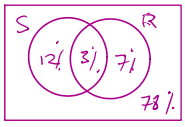
General case: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$



$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$$

Example - Error Types: In a software testing run:

- P(syntax error) = 0.15
- P(runtime error) = 0.10
- P(both) = 0.03



P(syntax OR runtime error) = 0.15 + 0.10 − 0.03 = 22%

P(no error) = 100% − 22% = 78%

Rule of Product (Multiplication Rule)

For independent events: $\boxed{P(A \cap B) = P(A) \times P(B)}$

for all events dependent or not

Independent events:

$P(B|A) = P(B)$

$$\boxed{P(A \cap B) = P(A) \times P(B|A)}$$

Example - Multiple Component Failure: A system has 3 independent hard drives, each with reliability 0.95.

- $P(all\ 3\ work) =$ $P(S_1) \times P(S_2) \times P(S_3) = (0.95)(0.95)(0.95) = 85.7\%$
- $P(at\ least\ one\ fails) =$ $1 - P(\text{no fail}) = 1 - 0.857 = 14.3\%$

Example - Hash Collisions: If a hash function distributes uniformly over 1000 buckets, what's the probability that 3 independent keys all hash to different buckets?

$$\frac{1000 \times 999 \times 998}{1000 \times 1000 \times 1000} = \frac{P(1000, 3)}{1000^3} = \text{No collision} = 99.7\%$$

Mutually Exclusive vs Independent Events

Critical Distinction:

Mutually Exclusive: A and B cannot both occur

- $P(A \cap B) = 0$
- Example: A packet is TCP OR UDP (not both)

Independent: Occurrence of A doesn't affect probability of B

- $P(A \cap B) = P(A) \times P(B)$ OR $P(B|A) = P(B)$
- Example: Errors in separate network links

Important: If A and B are mutually exclusive and both have positive probability, they CANNOT be independent!

Computing Example: File access permissions: Read (R), Write (W), Execute (X)

- Events "has R" and "has W" are independent (can occur together or separately)
- Events "access granted" and "access denied" are mutually exclusive

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Conditional Probability

Definition

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

This reads as "probability of A given B has occurred"

$$P(B|A) = \frac{P(A \cap B)}{P(A)} \qquad P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Example - Spam Filtering:

$P(\text{spam} \mid \text{"free"}) = \frac{P(\text{spam} \cap \text{free})}{P(\text{free})} = \frac{25\%}{40\%}$

- P(contains "free") = 0.40
- P(spam) = 0.30
- P(contains "free" AND spam) = 0.25

$P(\text{free} \mid \text{spam}) = \frac{P(\text{free} \cap \text{spam})}{P(\text{spam})} = \frac{25\%}{30\%} = 83.3\%$

P(spam | contains "free") = $\frac{0.25}{0.40} = 0.625$

So 62.5% of emails containing "free" are spam.

$P(A \cap B) = P(B \cap A)$

Properties of Conditional Probability

1. **Independence check:** Events A and B are independent if and only if $\boxed{P(A|B) = P(A)}$
2. **Multiplication Rule (General):** $\boxed{P(A \cap B) = P(A|B) \times P(B) = P(B|A) \times P(A)}$

Example – sampling without replacement: From a deck of standard cards we pick two cards without replacement. Find P(both Kings):

$P(K_1) = \frac{4}{52}$

$P(K_1 \cap K_2) = P(K_1) \times P(K_2 \mid K_1) = \frac{4}{52} \times \frac{3}{51}$

$P(K_2 \mid K_1) = \frac{3}{51}$

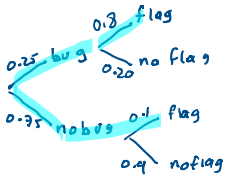
Example - Sequential Access: A database has 1000 records: 600 are "active" and of those, 200 are "premium".

$P(\text{premium} \mid \text{active}) = \frac{P(\text{both})}{P(\text{active})} = \frac{200}{600} = \frac{1}{3}$

- P(active) = $\frac{600}{1000} = 0.6$
- $P(premium \mid active) = 200/600 = 1/3$
- $P(active\ AND\ premium) = (\frac{1}{3}) \times 0.6 = 0.2$

$\frac{200}{1000} = 0.2$

Computing Example - Bug Detection: A code review process:



- P(code has bug) = 0.25
- P(reviewer flags bug | code has bug) = 0.80 (true positive)
- P(reviewer flags bug | no bug) = 0.10 (false positive)

P(reviewer flags code AND bug exists) = P(flags | bug) × P(bug) = 0.80 × 0.25 = 0.20

$0.10 \times 0.75 = 7.5\%$

	Bug	No bug	
Gets flagged	20%	7.5%	27.5%
Not flagged	5%	67.5%	72.5%
	25%	75%	100%

$(0.25 \times 0.8) + (0.75 \times 0.1)$

27.5%

Find P(reviewer does not flag bug | code has bug)

$= \frac{5\%}{25\%} = 20\%$

$P(\text{no flag} \mid \text{has bug}) = \frac{P(\text{no flag} \cap \text{bug})}{P(\text{bug})}$

4. Law of Total Probability

The Law

If $B_1, B_2, \ldots, B_n$ form a partition of the sample space (mutually exclusive and exhaustive), then:

$$P(A) = P(A|B_1)P(B_1) + P(A|B_2)P(B_2) + \ldots + P(A|B_n)P(B_n)$$

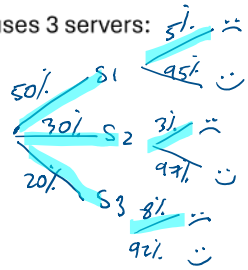
In the above example, what percent of the times the code gets flagged?

$$P(\text{flag}) = P(\text{flag} \mid \text{bug}) \times P(\text{bug}) + P(\text{flag} \mid \text{no bug}) \times P(\text{no bug})$$

$$(0.8 \times 0.25) + (0.10 \times 0.75) = 27.5\%$$

Computing Example - Server Response Time: A web application uses 3 servers:

- 50% of requests go to Server 1, where 5% timeout
- 30% of requests go to Server 2, where 3% timeout
- 20% of requests go to Server 3, where 8% timeout



What's the overall probability a request times out?

$$\begin{aligned} P(timeout) &= P(timeout|S1)P(S1) + P(timeout|S2)P(S2) + P(timeout|S3)P(S3) \\ &= (0.05)(0.50) + (0.03)(0.30) + (0.08)(0.20) = 0.025 + 0.009 + 0.016 \\ &= 0.050 \; or \; 5\% \end{aligned}$$

5% x 35% = 1.75%

1% x 25%

Computing Example - Software Testing: A program processes three types of inputs:

0.02 x 0.40 = 0.8%

- Type A (40% of inputs): 2% cause crashes
- Type B (35% of inputs): 5% cause crashes
- Type C (25% of inputs): 1% cause crashes

	A	B	C	
crash	0.8%	1.75%	0.25%	2.8%
no crash	39.2%	33.25%	24.75%	97.2%
	40%	35	25	100

P(crash) = (0.02)(0.40) + (0.05)(0.35) + (0.01)(0.25) = 0.008 + 0.0175 + 0.0025 = 0.028 or 2.8%

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5. Bayes' Theorem

$$The\ Theorem: P(A|B) = \frac{[P(B|A) \times P(A)]}{P(B)}$$

If we know P(B|A) this formula helps us find P(A|B)

Or using the law of total probability:

$$P(A|B) = \frac{[P(B|A) \times P(A)]}{[P(B|A)P(A) + P(B|A')P(A')]}$$ = Law of total probability for B

Why Bayes is Crucial in Computing

Bayes' theorem lets us "reverse" conditional probabilities - crucial for:

- Machine learning (Naive Bayes classifiers)
- Spam filtering
- Medical/diagnostic systems
- Network intrusion detection

Computing Example - Email Spam Filter:

$$P(\overbrace{SPAM}^{A} | \overbrace{"winner"}^{B}) = \frac{\overbrace{P(winner|spam)}^{0.60} \times \overbrace{P(spam)}^{0.30}}{P(winner)} = \frac{0.6 \times 0.3}{0.215} = \boxed{83.7\%}$$

$$P(winner) = P(winner|spam) \times P(spam) + P(winner|legit) \times P(legit)$$

$$(0.6 \times 0.3) + (0.05 \times 0.70)$$

$$= 0.215$$

Given:

- P(spam) = 0.30 (prior probability)
- P(legitimate) = 0.70
- P(contains "winner" | spam) = 0.60
- P(contains "winner" | legitimate) = 0.05

Find: P(spam | contains "winner")

Using Bayes:

$$P(spam \mid "winner") = \frac{[P("winner"|spam) \times P(spam)]}{[P("winner"|spam)P(spam) + P("winner"|legit)P(legit)]}$$

$$= \frac{(0.60 \times 0.30)}{[(0.60 \times 0.30) + (0.05 \times 0.70)]} = \frac{0.18}{[0.18 + 0.035]} = \frac{0.18}{0.215} \approx 0.837$$

So if an email contains "winner", there's an 83.7% probability it's spam.

Computing Example - Network Intrusion Detection:

Given:

- P(attack) = 0.001 (attacks are rare: 0.1% of traffic)
- P(IDS alerts | attack) = 0.95 (true positive rate)
- P(IDS alerts | normal) = 0.02 (false positive rate)

Find: P(attack | IDS alerts)

P(attack | alert) =

Alternative solution:

	attack	normal	
IDS alerts	0.00095	0.01998	0.02093
No allert	0.00005	0.97902	0.97907
	0.001	0.999	1

$$P(attack\ |Alert) = \frac{0.00095}{0.02093} = 4.5\%$$

Solution using Bayes' formula:

$$P(attack|IDS\ alerts) = \frac{0.95\times0.001}{(0.95\times0.001+0.02\times0.999)} = 4.5\%$$

Even with a 95% detection rate, only 4.5% of alerts are actual attacks! This shows why false positive rates matter so much in rare-event detection.

Key Insight: The base rate (prior probability) heavily influences the posterior probability. Rare events remain unlikely even with positive test results.

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6. Random Variables

Definition

A **random variable** is a function that assigns a numerical value to each outcome in a sample space.

We distinguish:

- **Discrete random variables:** Countable values (often integers)
- **Continuous random variables:** Uncountable values (we'll focus on discrete today)

Computing Example - Packet Loss: Let X = number of packets lost in a transmission of 10 packets

- X can take values {0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10}
- X is a discrete random variable

Probability Mass Function (PMF) $\sum_{i=1} P(X=i) = 100\%$

x	$P(x)$
0	10%
1	2%
2	8%
⋮	⋮
10	1%

For a discrete random variable X, the PMF is: **P(X = x)** or **p_x(x)**

Properties:

1. 0 ≤ P(X = x) ≤ 1 for all x
2. Σ P(X = x) = 1 (sum over all possible values)

Computing Example - Database Queries: Let X = number of database queries that succeed out of 3 attempts, where each has success probability 0.8.

x	$P(X = x)$
0	$C(3,0)(0.8)^0(0.2)^3 = 0.008$
1	$C(3,1)(0.8)^1(0.2)^2 = 0.096$
2	$C(3,2)(0.8)^2(0.2)^1 = 0.384$
3	$C(3,3)(0.8)^3(0.2)^0 = 0.512$

$(0 \times 0.008) + (1 \times 0.096) + (2 \times 0.384) + (3 \times 0.512)$

$\mu = E[X] = 2.4$

Note: These probabilities sum to 1.000

Expected Value (Mean)

$$\mu = E[X] = \Sigma\, x \cdot P(X = x) \text{ (sum over all possible values of x)}$$

The expected value is the long-run average value of the random variable.

For the database example: E[X] = 0(0.008) + 1(0.096) + 2(0.384) + 3(0.512) = 0 + 0.096 + 0.768 + 1.536 = 2.4 On average, 2.4 out of 3 queries succeed.

Computing Example - Server Load: Let X = number of active connections to a server

Connections (x)	P(X = x)	$x^2 P(x)$
0	0.10	$0^2 \times 0.1 = 0$
1	0.25	$1^2 \times 0.25 = 0.25$
2	0.35	$2^2 \times 0.35 = 1.4$
3	0.20	$3^2 \times 0.2 = 1.8$
4	0.10	$4^2 \times 0.10 = 1.6$
		$\Sigma x^2 P(x) = 5.05$

$E[X] = \mu = 0 \times 0.10 + 1 \times 0.25 + 2 \times 0.35 + 3 \times 0.20 + 4 \times 0.10 = 1.95 = \mu$

E[X] = 0(0.10) + 1(0.25) + 2(0.35) + 3(0.20) + 4(0.10) = 0 + 0.25 + 0.70 + 0.60 + 0.40 = 1.95 connections on average

$var = 5.05 - 1.95^2 = 1.2475$

$stdev = \sqrt{1.2475} = 1.12$

Variance and Standard Deviation

Variance: $Var(X) = E[(X - \mu)^2] = E[X^2] - (E[X])^2 = \boxed{\sum x^2 P(x) - \mu^2}$

Standard Deviation: $\sigma = \sqrt{Var(X)}$

Average number of active connections : 1.95 ± 1.12

2 ± 1

These measure the spread or dispersion of the random variable.

For the server load example: $E[X^2] = 0^2(0.10) + 1^2(0.25) + 2^2(0.35) + 3^2(0.20) + 4^2(0.10) = 0 + 0.25 + 1.40 + 1.80 + 1.60 = 5.05$

$$Var(X) = 5.05 - (1.95)^2 = 5.05 - 3.8025 = 1.2475$$

$$\sigma = \sqrt{1.2475} \approx 1.12\ connections$$

Interpretation: The typical deviation from the average load of 1.95 connections is about 1.12 connections.